

8(a)(i)	receptor ;	1
8(a)(ii)	intermediate neurone ; A relay neurone / motor neurone	1
8(a)(iii)	Schwann ;	1
8(b)	<i>any four from:</i> <i>differences</i> <i>myelinated neurone</i> 1 faster (impulse transmission) ; ora <i>explanation</i> 2 myelin (sheath), insulates (the axon) / prevents movement of ions ; 3 nodes of Ranvier / gaps with no myelin sheath, have voltage-gated channels or nodes of Ranvier / gaps with no myelin sheath, allow movement of ions ; 4 depolarisation / action potential, occurs only at nodes of Ranvier / gaps with no myelin sheath; ora 5 action potentials jump from node to node / saltatory conduction ; ora 6 longer local circuits / local circuits occur between nodes ; ora	4
8(c)(i)	1 A voltage-gated <u>and</u> B ligand-gated / not voltage-gated or A opens in response to, depolarisation / arrival of an action potential <u>and</u> B opens in response to binding of, acetylcholine / neurotransmitter ; 2 A for calcium ions <u>and</u> B for sodium ions ;	2

Question	Answer	Marks
8(c)(ii)	any four from: 1 no Ca^{2+} enter (presynaptic, knob / neurone) ; 2 vesicles do not, move towards / fuse with, presynaptic membrane ; 3 no, ACh / neurotransmitter, released or no exocytosis of, ACh / neurotransmitter ; 4 ACh / neurotransmitter, does not bind to, receptor / ion channel B ; 5 Na^+ does not enter postsynaptic neurone / ion channel B does not open ; 6 postsynaptic membrane not depolarised or no action potential generated or impulse not transmitted ;	4
8(c)(iii)	any two from: 1 inside more negative (charge) or membrane hyperpolarised / membrane potential decreases / potential difference increases ; 2 harder to / cannot, reach threshold ; 3 no depolarisation of membrane / harder to depolarise membrane or no / fewer / harder to generate, action potentials / impulses ;	2